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Tariqul Islam , Md. Zaidul Islam Sarker , ABM Helal Uddin , Kamaruzzaman Bin Yunus , Reddy Prasad D.M. , Md. Abdur Rashid Mia & Sahena Ferdosh

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Kamlet Taft Parameters: A Tool to Alternate the Usage of Hazardous Solvent in Pharmaceutical and Chemical Manufacturing/ Synthesis - A Gateway towards Green Technology

Tariqul Islam [1], Md. Zaidul Islam Sarker [1]*, ABM Helal Uddin [1], Kamaruzzaman Bin Yunus [2], Reddy Prasad D.M. [3], Md. Abdur Rashid Mia [1] and Sahena Ferdosh [2]

[1] Faculty of Pharmacy, International Islamic University Malaysia, 25200 Kuantan, Pahang, Malaysia
[2] Faculty of Science, International Islamic University Malaysia, 25200 Kuantan, Pahang, Malaysia
[3] Petroleum and Chemical Engineering Programme area, Universiti Technology Brunei, Tungku Highway, Gadong BE1410, Brunei Darussalam

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Abstract: In chemical synthesis, the uses of solvents are inevitable, and it has been a practice since the inception of the process. With time it gets into the notice that most of the essential solvents have a great potential to cause acute health hazards, which include health-related complications as severe. There was a continuous urge to getting some solution to overcome this problem without compromising the quality of the synthesis process. Kamlet Taft parameter (K-T) is such a tool that can provide detailed information about the solvent characteristics related to the synthesis process thus enabling the selection of the solvent with a lower hazardous index compared with the high one. It has also contributed significantly to tackle environmental hazards by replacing the hazardous dipolar solvent in synthetic chemistry. A solvent pair mixture-Hydrogen bond donor and Hydrogen bond acceptor- is used to replace the routine hazardous solvents. This review aimed at the systematic highlight of the K-T parameters, which will be useful for the selection of suitable solvents for different chemical synthesis, specifically giving importance to drug synthesis. The review will provide insight into the different applications of K-T parameters to ease the comparison of these applications. It will also guide to estimate the solvent effects in synthesis, crystallization, and formulations of pharmaceuticals and biopharmaceuticals with systematic classification, analysis, and comparison of the reported work.

Key words: Kamlet Taft parameters; solvent characteristics; hazardous index; hazardous dipolar aprotic solvent; solvent-pair mixture.

Introduction

Many hazardous solvents are being used in active pharmaceutical ingredients (API) and non-API chemical processing nowadays. Recently, Green Chemistry Institute (GCI), European academic chemical societies (EACS), American Chemical Society (ACS), and global pharmaceutical industrial consortia are actively studying to replace those hazardous solvents with alternative green solvents [1-3]. Already, the Innovative Medicines Initiative (IMI-CHEM21) [1], GlaxoSmith Kline-GSK [3], and the ACS-GCI Pharmaceuticals Roundtable [4, 5], prepared solvent selection guidelines. GSK revealed a statistical report that sol-

*Corresponding authors (Md. Zaidul Islam Sarker)
E-mail: < zaidul@iium.edu.my >

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vents are used around 80-90 % of the non-aqueous materials during the production of active pharmaceutical ingredients. This report indicates, as a primary method, that the net outcome of the processing of ingredients with a significant amount of waste is known. Ultimately, these impurified materials are being used in discovery, improvement, and even in processing of pharmaceutical ingredients with a large scale of production [6, 7]. Another report stated that at least 50 % solvent of total ingredients is required in the manufacturing of pharmaceuticals and chemicals [8].

However, the maximum use of green solvents can minimize harmful environmental effects arising from manufacturing industries, and it could be the most effective way to limit the use of hazardous solvents [9-12]. Some pharmaceutical companies are using their self-prepared solvent selection guide to avoid wastage and unexpected toxicity [6, 13]. Total solvents have been classified into four groups: recommended, problematic, hazardous, and highly hazardous solvents [1]. However, hazardous dipolar aprotic solvents are most effective in the various API operations such as the solubilization process [2, 14, 15], crystallization steps [4, 16], formulations, laboratory analysis, and reactions [12].

Such solvents have high Kamlet-Taft polarity (polarity > 0.6), high hydrogen bond accepting capacity (basicity > 0.6), and low hydrogen bond donating capability (acidity ~ 0) that gives strong fundamental interaction with the active materials [2, 4, 14, 15]. But, pure recommended solvents (Example: Methanol, Ethanol, Acetone) and most of the solvent-pair mixtures (Example: H_2O-EtOH, MeOH-EtAc) don't provide hazardous activities. On the other hand, the recommended pure single solvent can't show the same chemical interaction as like hazardous solvents. So, the solvent-pair mixtures can be used in the replacement of hazardous ones because most of the solvent-pair mixtures carry the same chemical interaction as hazardous solvents [5].

Some specific indicators or dyes are utilized in the analysis of the K-T parameters. This indicator-solvent mixture gives the desired wavelength against the indicator absorbance using a UV-VIS spectrophotometer that helps to calculate the solvent activities (di-polarity, basicity, and acidity) in solution [17]. So, this method can be applied to preparing a solvent mixture guide for total pharmaceutical and chemical ingredients for the hazardous solvent replacement.

Solvent classification

A universal principle is often pursued when developing the classification of solvents to be used in pharmaceutical and chemical factories around the world. The color-coding system is a universally applied 'traûc light' scheme with the remark on particular every solvent to the conditions executed by each organization. Where Pûzer considered a solvent as 'usable' and as 'some issues' is stated by GSK, besides that, Sanofi recommended as 'substitution advisable'. Already, some organizations published their solvent guides as per their requirement, for example, Pfizer, GSK, and Sanofi [13, 18, 19].

Recently, solvent guidelines of GSK, Pfizer, and Sanofi have been well developed, and all solvents were presented in the three comments or classes [20]. GSK already has prepared a solvent selection guide for chemists and researchers that published through the medicinal chemistry and as a tool of Pûzer [21, 22]. On the other hand, another report stated that the existing solvent selection guides provide only quasi-quantitative data about green solvents [23].

Prat *et al.*[1, 6] stated that solvents could be classified into four categories according to CHEM21 guideline: solvent details demonstrated in Table 1. Apart from this, Prat *et al.*[1] described solvent classification as the CHEM21 conventional and unconventional solvent selection guide [1, 23]. Byrne *et al.* [24] also published a report regarding the solvent activities in the chemical processing of classified hazardous and referred solvent according to GSK solvent guidelines.

In 2019, 'the committee for human medicinal products' of the European Medicines Agency republished a solvent classification report (ICH, International Conference on Harmonization guideline, Q3C (R6)) which depends on the residual solvents in pharmaceutical products [25]. For example, Class 1 solvents, to be avoided due to undesirable toxicity or their adverse impact on the

Table 1. Solvent categories, according to the CHEM21 solvent selection guideline [1, 6]

Category	Condition	Solvent
Recommended (or preferred)	Require solvent test firstly then exercise	Water, Ethanol, i-PrOH, n-BuOH, EtOAc, Sulfonale, i-PrOAc, anisole, n-BuOAc
Recommended or Problematic?	Need to follow the recommended and problematic solvent's condition	Methanol, alcohol, acetone, ethylene t-BuOH, benzyl, glycol, MEK, MIBK, cyclohexane, MeOAc, Acetic acid, Ac_2O
Problematic	Possible in use in Lab or Kilolab, but require measurement in pilot plant or large scale production	Me-THF, heptane, Me-cyclohexane, toluene, xylenes, chlorobenzene, acetonitrile, DMPU, DMSO
Problematic or Hazardous?	Need to follow the problematic and hazardous solvent's condition	MTBE, THF, cyclohexane, DCM, formic acid, pyridine
Hazardous	Very strong constrains in the scale-up use, even have priority to substitution during use	Diisopropyl ether, 1,4-dioxane, DME, pentane, hexane, DMF, DMAc, NMF, methoxy-ethanol, TEA
Highly hazardous	Always should avoid, even in the research lab	Diethyl ether, benzene, chloroform, CCl_4, DCE, nitromethane, CS_2, HMPA

Abbreviation: n-BuOH=1-butanol, EtOAc=ethyl acetate, iPrOAc=isopropyl acetate, n-BuOAc=n-butyl acetate, t-BuOH=tert-butanol, MeOAc=methyl acetate, Ac_2O=acetic anhydride, MEK=methyl ethyl ketone, MIBK=methyl isobutyl ketone, DMPU=dimethylpropyleneurea, MTBE=methyl tert-butyl ether, DCM=dichloromethane, TEA=triethylamine, CCl4=carbon tetrachloride, DCE=dichloroethane, CS_2=Carbon disulfide, HMPA=hexamethylphosphoramide and see table 2 for others abbreviation

environment and Class 2 solvents should limit in pharmaceutical products due to their intrinsic toxicity. The third type of solvents is class 3 solvents that are recommended to use in pharmaceuticals due to the low toxic potential. These all organizations categorized solvents just according to their requirements. But, they did not establish a solvent guide yet for the replacement of hazardous solvents and even for specific pharmaceutical ingredients.

The solubility of solvent mixture (HBD-HBA)

The solubility properties of the solvent mixture (hydrogen bond donor (HBD) - hydrogen bond acceptor (HBA)) in the pharmaceutical ingredient are identified using K-T parameters [2]. In the replacement of hazardous solvent, K-T parameters can easily explain solvent activities and solubility in API [5]. Most of the hazardous solvents have a fixed dipole moment, low reactivity, and high polarizability that unable to donate unstable hydrogen atoms to build a strong hydrogen bond [26]. Maximum solvent-pair mixtures show the same properties in the reaction as hazardous dipolar solvent [5]. On the other hand, the solubility of a pharmaceutical ingredient in a solvent or their mixture is also calculated by the various thermodynamic and kinetic solubility determination methods [27].

Pharma modified UNIFAC [28], PC-SAFT (Perturbed-Chain Statistical Associating Fluid Theory) [29], novel method COSMO-RS [30], many methods have been developed to measure the co-efficient behaviour and solubility of a solid ingredient in a solvent or solvent mixture. Then, many data like enthalpy, temperature, molar volume, molecular weight, and solubility data of active ingredients and solvents are also required to measure activity co-efficient in utilizing the mentioned theories. Working composition and GSK safety, health, environment, and waste score are required to pre-

pare a ranking of solvent-pair mixtures. The working composition is ascertained at the maximum level of solubility [5, 31]. Necessary some solubility parameters, K-T parameters, GSK2016 scores of safety, health, environment, and waste of various solvents have been tabulated (Table 2) from literature.

Kamlet-Taft parameter

Kamlet-Abbound-Taft (KAT) parameter was introduced to identify the solvent characteristics and mechanism in a solution [34]. The KAT parameter can determine the polarity (π*) [35], accepting capacity of hydrogen-bond (β) [36], and donating ability of hydrogen-bond (α) [37] of a solvent. These three parameters are usually determined from the absorbance of the normalized UV-VIS spectrophotometer. At the beginning of UV analysis, a specific ratio of solvatochromic indicators or dyes is used in the solution; for that, it is called K-T solvatochromic parameters [17, 38]. Recently, the original method of Kamlet and Taft has been modified. Some modified methods are also being applied to measure α, β, π* parameters like K-T Solvatochromic Parameters Using COSMO-RS [34], K-T Parameters using Solvate Ionic Liquids [39-41], Solute-Centric K-T Solvatochromic Parameter [42].

Table 2. Solubility parameter at 25°C [32], Kamlet-Taft parameters [33, 54], GSK safety, health, environmental, and waste scores [3] for various solvents

Name		δ MPa$^{0.5}$	Kamlet-Taft α	β	π*	*GSK Ranking Scores Safety	Health	Environmental	Waste
Dipolar	DMF	24.9	0.0	0.7	0.9	9.0	2.4	6.3	4.6
Aprotic	DMSO	26.7	0.0	0.8	1.0	6.7	7.9	6.9	4.6
	DMAc	22.8	0.0	0.8	0.9	9.0	2.6	7.7	4.7
	ACN	24.4	0.2	0.4	0.7	7.7	5.9	8.9	2.8
	NMP	23.0	0.0	0.7	0.9	9.0	3.0	7.7	4.4
	Pyr	21.8	0.0	0.7	0.9	8.5	4.0	4.6	4.0
Water	Water	47.8	1.2	0.5	1.2	8.9	9.5	8.9	3.7
Alcohol	EtOH	26.5	0.9	0.8	0.6	7.7	8.9	6.7	4.2
	i-PrOH	23.2	0.8	0.5	0.5	6.9	7.7	7.5	4.4
	n-PrOH	24.6	0.8	0.9	0.5	8.9	8.4	6.3	4.1
	MeOH	29.6	1.1	0.7	0.7	7.1	4.9	8.4	4.0
Ketone	Ace	19.9	0.1	0.5	0.6	6.0	7.7	7.7	3.3
	CHN	19.6	0.0	0.6	0.7	8.5	6.5	7.1	7.4
	CPN	22.1	0.0	0.6	0.8	8.9	6.5	7.1	7.4
Ester	i-PrAc	17.6	0.0	0.4	0.5	7.7	7.7	6.7	5.7
	EtAc	18.2	0.0	0.5	0.5	7.1	8.4	6.7	4.9
	MeAc	18.7	0.0	0.4	0.5	6.0	7.0	9.0	3.0
Lactone	GVL	21.3	0.0	0.6	0.9	9.5	5.3	7.7	8.4
	GBL	26.3	0.0	0.5	0.9	9.5	5.3	7.7	8.4
Ether	THF	19.5	0.0	0.6	0.6	4.9	5.9	5.2	3.5
	DI	20.5	0.0	0.4	0.5	4.9	3.5	5.7	2.9
	DEET	15.9	0.0	0.5	0.2	3.5	6.3	3.9	3.5

Abbreviation: DMSO = dimethyl sulfoxide, ACN = acetonitrile, Pyr = pyridine DMF = dimethyl formamide, DMAc = dimethyl acetamide, NMP = *N*-methyl-2-pyrrolidone, EtOH = ethanol, i-PrOH = isopropanol, n-PrOH = 1-propanol, MeOH = methanol, Ace = acetone, CHN = cyclohexanone, CPN = cyclopentanone i-PrAc = isopropyl acetate, EtAc = ethyl acetate MeAc = methyl acetate, GVL = γ-valerolactone, GBL = γ-butyrolactone, DEET = diethyl ether THF= tetrahydrofuran, DI = 1, 4-dioxane and table 3* for solvent ranking

Conditions in the Kamlet Taft parameters analysis

Double beam UV-VIS spectrophotometer, 0.2 nm resolution, and 1 cm path length of quartz cells are applied to determine UV-VIS absorbance of the solvent with indicator. A temperature controller can be used to maintain the temperature of the UV spectrophotometer at 25 ± 0.1°C. Then, the indicator in a solvent or a solvent mixture is appropriately stirred before analysis [17]. Indicators are organized by weighing using a microbalance with an uncertainty of ±1×0.1 mg [17]. Even, Specific concentration of indicator is maintained so that variety of UV absorption could be limited within 0.5-1.2 [42]. Duereh *et al.* [42] followed that the wavenumber interval of UV analysis was 0.2 nm, and the uncertainty of the wavelength was ± 0.8. Besides that, the repetition of every UV-Vis spectra has been performed by a minimum of 3 scans. Then, the K-T parameters of the solvents are calculated by applying some specific equations. Wavelength standard (Holmium (III) oxide liquid in Perchloric acid) is used to calibrate with an uncertainty (2σ) in maximum absorption wavelength (λ_{max}) of ± 0.4 nm in the measurement of the spectrophotometer [42, 43].

Theoretical reviews in the calculation of α, β, and π*

The highest wavelength value from the absorbance of the indicator in the solvent or solution is selected to obtain the K-T parameters. Almost the same wavelength is observed when three scannings were performed. The indicators are N, N-dimethyl-4-nitroaniline 98 % (Indicator-1 for π* value), 4-nitroaniline 99 % (Indicator-2 for β value) and Reichardt's dye 90 % (Indicator-3 for α value). Usually, the concentration of indicator (Indicator I & 2) in the solvent or solution is 0.03 mM to 0.05 mM and 0.1mM for indicator 3 [17]. Specific concentration is maintained so that variety of UV absorption could be limited within 0.5-1.2 [42]. These parameters were measured using the equations (eq. 1 to 3) of the original technique of Kamlet Taft [35-37, 44].

Indicator 1: Polarity (π*) = (28.10 - V_{max1}) / 3.52 (eq. 1)

Indicator 2: Basicity (β) = (0.984V_{max1} + 3.49 - V_{max2}) / 2.759 (eq. 2)

Indicator 3: Acidity (α) = (1.318V_{max1} - 47.7 + V_{max3}) / 5.47 (eq. 3)

Firstly, the wavelength (nm) is converted into wavenumber (Vmax/V) by simply dividing 10000000. Then, the wavenumber is converted into KiloKaiser unit (1KK = 1000 cm^{-1} = 10000/ λ_{max} (nm) require to calculate the equations.

Recently, the Kamlet-Taft method has been implemented using the four equations (eq. 4 to 7) to estimate the K-T parameters, in view, that K-T basicity indicates a linear relationship with Hunter's basicity [45].

Indicator 1: Polarity (π*) = (28.10 - V_{max1}) / 3.52 (eq. 4)

Indicator 2: Basicity (β) = 0.358 X (31.10 - V_{max2}) - 1.125 X π* (eq. 5)

Indicator 3: Acidity (α) = 0.0649 X E_T (30) - 2.03 - 0.72 X π* (eq. 6)

Here, E_T, $Kcal.mol^{-1}$ = 28591.5/λ_{max3} (nm) (eq. 7)

Indicators are used to measure K-T parameters, as their UV ranges are higher than the current limit of UV-Vis-310 nm in the cyclic ketones. To average the results, two/three indicators for every K-T parameter are advantageous that gives accurate result than other results where a single indicator is used [42, 43, 45, 46].

In 1997, A.S. Labban, and Y. Marcus modified the Kamlet-Taft method. This Marcus method held seven equations using seven indicators to calculate the K-T parameters [47]. As well as, Duereh *et al.* [43] published a report where six equations and six indicators had been applied [43].

Indicator (1): N, N - dimethyl-3-nirtoaniline, π* = 0.452 (25.52 - V) (eq. 8)

Indicator (2): N, N-dimethyl-4-nitroaniline, π* = 0.314 (27.52 - V) (eq. 9)

Indicator (3): 4-nitroaniline, β = 0.358 (31.10 - V)-1.125 π* (eq. 10)

Indicator (4): (tetramethylethylenediamine) (acetylacetonato) copper (II) perchlorate:

β = 0.358 (18.76 - V) (eq. 11)

Indicator 5: 4-aminoacetophenone, β = 0.245 (35.25 - V) (eq. 12)

Indicator 6: Cis-bis (1,10 phenanthroline) dicyanoiron (II)

$\alpha = 0.375\ (V - 15.636) - 0.45\ \pi^* + 0.72\ \beta$ (eq. 13)

Indicator 7: 2, 6-diphenyl-4-(2, 4, 6-triphenyl-1-pyridinio) phenoxide/phenolate

$\alpha = 0.0649\ E_T(30) - 2.03 - 0.72\ \pi^*$, $E_T(30) = 2.859V$ (eq. 14)

In the modification method, the concentration of indicators in the solvent or solution was 0.5 - 2.0 X 10^{-4} mol.dm^{-3}. Wavelength/absorbance has been measured against solvent blank by using UV-Vis spectrophotometer at room temperature, 25 ± 2ºC. They controlled separate two samples for each solvent-indicator combination, so that UV wavelength ranges may be restricted under 0.500 nm [47].

Application of K-T parameter in the synthesis

Methodology for replacing dipolar aprotic solvents has been used in API processing with solvent-pair mixtures shown in Figure 1. Firstly, API properties (such as molar volume, melting temperature, enthalpy of fusion) and solvent properties such as KT and solubility parameters have to estimate. Then, the API solubility parameter and API Kamlet-Taft parameter window have to define, and working composition ranges of solvent-pair mixtures are determined. After preparing the solvent ranking, the solvent mixtures are selected to replace hazardous solvents in API processing. This ranking expresses information regarding the

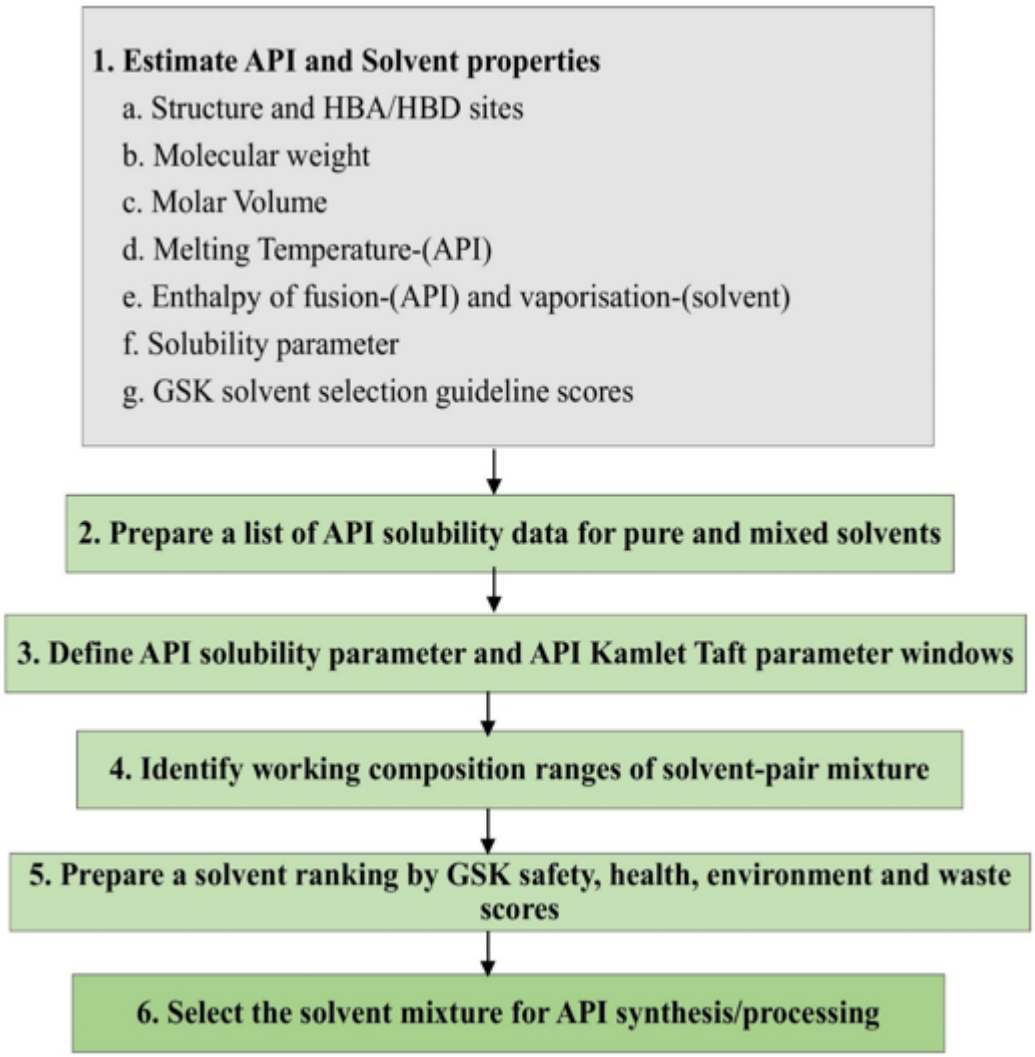


Fig. 1. Methodology for replacing dipolar aprotic solvents used in API synthesis/processing with safe hydrogen-bond donor (HBD) and hydrogen-bond acceptor (HBA) solvent-pair mixtures [5]

ability of the solvent to give the desired reaction with an API without hazardous activities [17]. The ranking can be ascertained from the minimum value of the GSK health and safety scores using the equation (15). Another way to determine the ranking of solvent-pairs, at first GSK scores is determined according to GSK guidelines in the targeted solvent mixture for specific API. Then, the ranking is prepared according to the equation (16) [5]. Colour code and score range are also used to prepare the ranking for the specific API shown in Table 3.

Composite score = {safety (S) x health (H)}$^{1/2}$ (eq. 15)

Composite score = {Safety (S) x Health (H) x Environmental (E) x Waste (W)}$^{1/4}$ (eq. 16)

Duereh *et al.* [5] revealed that the solvent-pair mixtures provide good paracetamol solubility, including the API solubility parameter range (16.3-30.5 MPa 0.5) and the API-KT parameter ranges ((polarity ≥ 0.35), (acidity ≤ 0.75), and (basicity ≥ 0.62)). Duereh *et al.*[42] had determined that higher solute-centric basicity, higher solute-centric polarity, and lower solute-centric acidity are (β_{SC} > 0.60), (π^*_{SC} > 0.63), and (α_{SC} < 0.63) respectively. In that study, the two monomers of this reaction were 4, 4-oxidianilin, and pyromellitic dianhydride that was investigated using a spectrophotometer to form the typical engineering plastic polyimide precursor, known as homogeneous poly (amic acid) [42]. In this synthesis, the mixture of tetrahydrofuran (THF) solvent: THF-methanol, THF-ethanol, THF-water were used in the replacement of hazardous solvent N-methyl-2-pyrrolidone [55, 56] and diethyl ether [56] using the Kamlet Taft method. The methodology of selecting a replacement solvent mixture for the synthesis of polymer precursor solutions via homogenous reaction is demonstrated in figure 2.

The other report expressed that cyclohexanone (CHN) and cyclopentanone (CPN) are possible to obtain easily from renewable biomass resources [48, 49] that time CHN and CPN had been established to use as biofuels and fine chemicals [50, 51]. Both CHN and CPN had also been applied for industrial processes as molecular solvents [52], and as poly (vinyl chloride) fractionation [53] because they have a high polarity (π^* ~ 0.78) of K-T, low acidity (α ~ 0), and high basicity (β ~ 0.54) [54]. The mixed solvent system of CPN and CHN: MeOH-CHN, EtOH-CHN, MeOH-CPN, EtOH-CPN have been studied to 12 APIs to consider the substitution of dipolar aprotic solvents [17]. Duereh *et al.* [5] prepared a list of 52 solvent-pair mixtures using Kamlet-Taft parameters and also applied it to 13 APIs. From that study, maximum solvent mixtures have been proposed to apply in the replacement of hazardous solvents. With time, many solvent-pairs have been studied for the specific API and non-API chemicals shown in Table 4.

Conclusion and future perspective

This review describes the overview of the recent progression of Kamlet-Taft parameters to measure the solvent activities in API and non-API chemical processing. The measurement procedure of Kamlet-Taft parameters using solvatochromic indicators or dyes is most efficient to calculate the K-T parameters in the pure and mixed solvent-pair system. Then, the original Kamlet-Taft method is modified that is being applied effectively now. Especially, the replacement of hazardous solvent using Kamlet-Taft parameters can be easily implemented that is well suited to sustainable and green chemistry that can minimize undesirable toxicity or their harmful effects on the environment. Such parameters can be anticipated in future time to be an exciting basic technology. Apparently, the Kamlet-Taft parameter can be a key tool to replace the hazardous dipolar aprotic solvents at the moment but need a more active study.

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Table 3. Colour code and score range for preparing a solvent ranking according to IMI-CHEM21 [1]

Colour code	Score range	Ranking status
Green highlight	7.00-10.00	Recommended solvent-pairs
Yellow highlight	4.00-7.00	Problematic solvent-pairs
Red highlight	0.00-4.00	Hazardous solvent-pairs

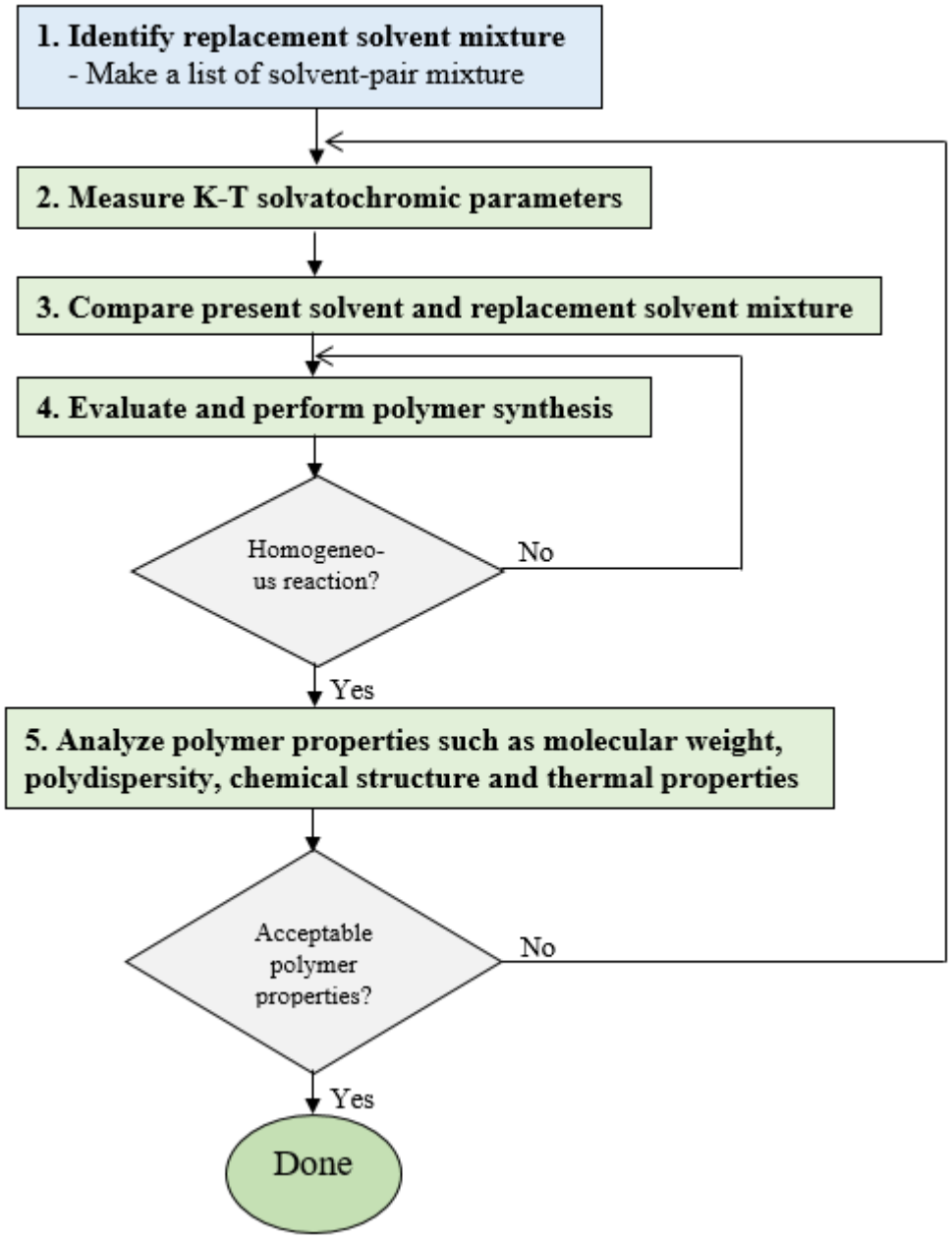


Fig. 2. Methodology of selecting replacement solvent mixture for the polymer synthesis [42]

Table 4. Application of solvent mixtures in API and non-API chemicals from literature

Ingredient/API	Solvent-pair mixture
[5] Ibuprofen, Paracetamol, Salicylic acid, Aspirin, Sulfamethoxy pyridazine, Benzoic acid, p-aminobenzoic acid, p-hydroxybenzoic acid, Naproxen, Temazepam, Niflumic acid, Piroxicam	(1) H_2O-GVL, H_2O with GBL, EtOH, iPrOH, t-BuOH, DMSO, ACN, NMP, Ace, DMF, MeOH, Pyr, THF, and DI (2) EtOH-GBL, EtOH with GVL, CPN, iPrAc, BuAc, CHN, EtAc, DMSO, MIBK, Anisole, ACN, NMP, MEK, DMF, Ace, Pyr, MTHF, THF and DI (3) MeOH-GBL MeOH with GVL, CPN, CHN, Anisole, iPrAc, EtAc, DMSO, BuAc, MIBK, ACN, Ace, MEK, NMP, DMF, Pyr, THF, MTHF and DI
[5] Sulfadiazine	H_2O-DMSO, H_2O-DI, EtOH-DMSO, MeOH-DMF, H_2O-GVL, EtOH-DMF, H_2O-tBuOH, MeOH-NMP, EtOH-ACN, H_2O -DMF, H_2O Pyr, EtOH-NMP, MeOH-DMSO, MeOH-Pyr, EtOH-Pyr, H_2O -NMP, H_2O-THF
[42] poly (amic acid) from 4, 4-oxidianilin and pyromellitic dianhydride	Ethanol-THF, Methanol-THF, H_2O-THF
[17, 43] K-T parameters analysis for the replacement of hazardous dipolar aprotic solvents	H_2OˆGBL, EtOHˆGBL H_2OˆGVL, MeOHˆGVL, EtOHˆGVL, MeOHˆGBL MeOH-CHN, EtOH-CHN, MeOH-CPN, EtOH-CPN

Note: Abbreviation of solvents have been presented in Table 1 and 2 and some solvent-pairs unable to reaction like, for Paracetamol: MeOH-Ace, MeOH-CPN, MeOH-EtAc, H_2O-ACN, H_2O-MeOH, H_2O-EtOH, MeOH-iPrAc, MeOH-GBL, MeOH-BuAc, H_2O-GBL, H_2O-iPrOH, MeOH-Ans, MeOH-DI; Ibuprofen: H_2O-MeOH; Naproxen and Piroxicam: H_2O-DMSO, H_2O-iPrOH, EtOH-DMSO, MeOH-DMSO, H_2O-GBL, EtOH-GBL, MeOH-GBL, H_2O-EtOH, H_2O-MeOH, MeOH-EtAC; Salicylic acid: H_2O-MeOH

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